

Scaffolding Low-Resource STEM Education with Large Language Models

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Abstract

Access to STEM education can be challenging for learners from low resource language backgrounds because English-medium instruction and limited localized technical resources can restrict understanding of specialized concepts. A Systematic Literature Review (SLR) and Systematic Systems Review (SSR) show that existing research and systems address multilingual translation, STEM terminology, educational scaffolding, and Large Language Model (LLM)-based learning, but these capabilities remain fragmented. This creates a practical problem for Burmese-speaking learners needing language and conceptual support, and a research problem concerning how these capabilities can be integrated into an educational information system. Grounded in Scaffolding Theory and the Cognitive Theory of Multimedia Learning, this study derives design principles for identifying STEM terminology, providing context-sensitive Burmese support, explaining concepts, and structuring learner assistance. These principles form the conceptual artefacts and guide the development of corresponding system artefacts for supporting Burmese-speaking learners in understanding specialized English STEM terminology and concepts.

Keywords: Educational Scaffolding, Large Language Models, Low Resource Languages, STEM Terminology, Multilingual Learning, Burmese Language

1. Search Coverage & Currency

A Systematic Literature Review (SLR) and Systematic Systems Review (SSR) were conducted to identify current research and technologies relevant to scaffolding STEM education using Large Language Models (LLMs) for learners from low resource language backgrounds. Within the Observe phase of the Information Systems Research (ISR) process used in this study, the reviews establish the theoretical and technological foundations for subsequent conceptual and system artefacts, consistent with the role of observation and systems development in IS research described by Nunamaker et al. (1990). The review was informed by the literature review typology of Paré et al. (2015), while study identification and screening were documented using PRISMA 2020 to support transparent and complete reporting (Page et al., 2021).

Scopus was used as the sole bibliographic database. Its multidisciplinary coverage was appropriate to the research scope because Scaffolding Low-Resource STEM Education with Large Language Models intersects artificial intelligence, information systems, education, STEM, computational linguistics, and

multilingual language technologies. The database also aligned with the study's keywords concerning LLMs, low resource languages, STEM education, educational scaffolding, multilingual learning, and translation. This breadth enabled literature from both the educational and technological dimensions of the research problem to be retrieved within a consistent search environment.

The search strategy operationalized the research scope through five interconnected concepts: LLMs and generative AI, education and learning, low resource and multilingual languages, educational or terminology support, and STEM. Synonyms and related terms were combined using Boolean operators and searched within titles, abstracts, and keywords. The final Scopus search query was:

TITLE-ABS-KEY(("large language model" OR "generative AI" OR "generative artificial intelligence" OR ChatGPT OR "language model*" OR "LLM") AND (education* OR learning OR student* OR learner* OR "educational technolog*" OR "intelligent tutoring system*") AND ("low-resource language*" OR "low resource language*" OR "under-resourced language*" OR "resource-poor language*" OR "linguistically diverse" OR multilingual OR "English language learner*" OR "non-native English speaker*" OR "Burmese" OR "Myanmar") AND (scaffold* OR "learning support" OR "conceptual explanation*" OR "educational support" OR "pedagogical support" OR "terminology support" OR "vocabulary support" OR translation OR "technical terminology") AND (STEM OR science OR technolog* OR engineering OR mathematic* OR "technical education" OR "technical subject*")) AND (LIMIT-TO (LANGUAGE,"English"))*

The search returned 188 records. An Open Access filter was applied to ensure that complete texts were available for detailed assessment, leaving 67 records for title and abstract screening. Studies were retained when their primary contribution was relevant to the intersection of LLMs or generative AI with multilingual or low resource language technologies, education, translation, terminology support, or STEM learning. Studies whose primary contributions were unrelated to the research problem, lacked a meaningful educational or multilingual contribution, or addressed general NLP applications without transferable relevance to the proposed educational artefacts were excluded.

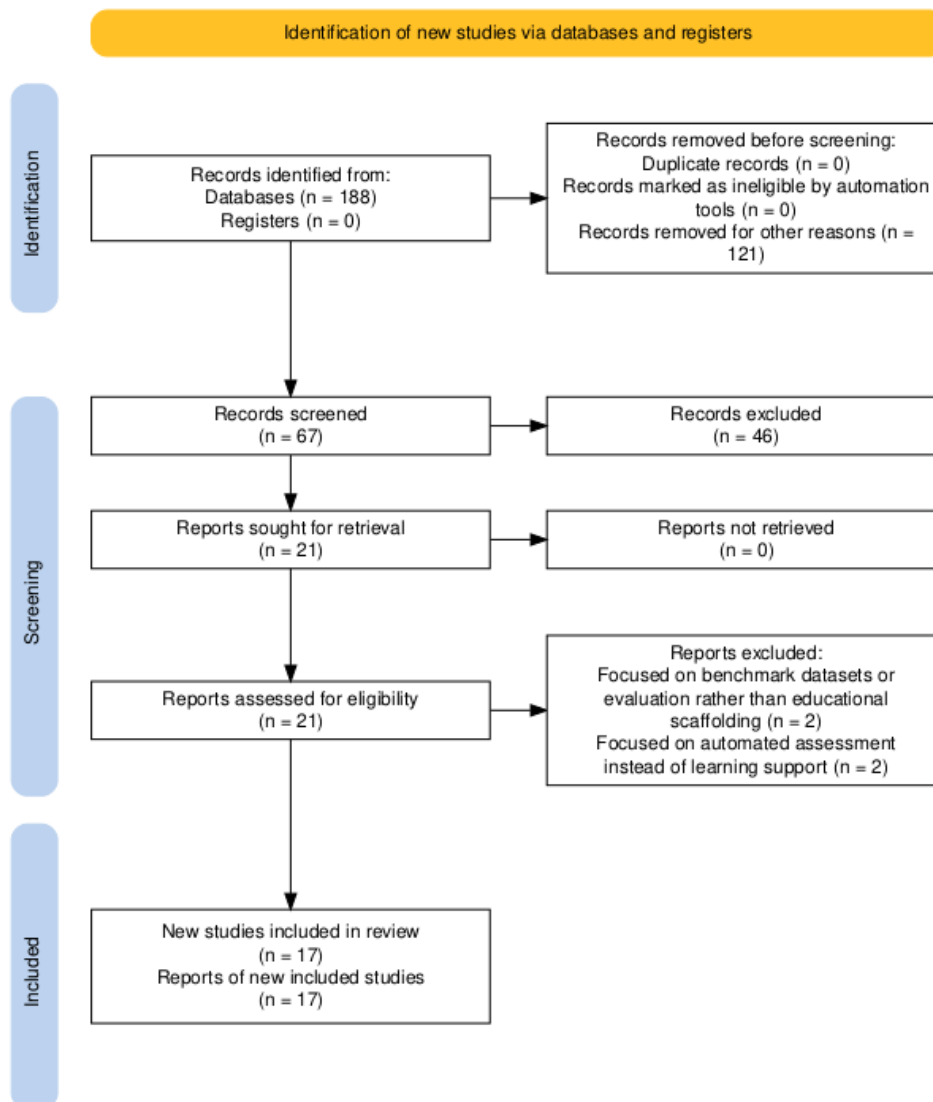
Following title and abstract screening, 21 reports were sought for retrieval and all 21 were retrieved and assessed for eligibility through full-text review. Full texts were evaluated for their contribution to multilingual or low resource language support, LLM-enabled educational support, terminology or translation, educational scaffolding, and knowledge relevant to conceptual or system artefact development. Four reports were excluded because their principal contributions concerned benchmark datasets or automated

assessment rather than educational scaffolding or learning support. Consequently, 17 studies were included in the final SLR and thematic synthesis.

The complete selection process is presented using the PRISMA 2020 framework in Figure 1. The diagram records each stage from identification through screening, eligibility assessment, and final inclusion, providing a transparent account of how the final evidence base was established.

Figure 1

PRISMA 2020 Study Selection Process



2. Systematic Literature Review (SLR)

The 17 studies retained after full-text screening were analyzed thematically to identify recurring findings relevant to LLM-supported STEM education in low resource language contexts. Four themes emerged: (1) low resource and multilingual language barriers, (2) LLM-based translation and terminology support, (3) educational scaffolding and personalized learning, and (4) system design and prompt engineering.

Together, these themes provide conceptual and technological knowledge for designing educational support for Burmese-speaking STEM learners.

2.1. Low Resource and Multilingual Language Barriers

Unequal linguistic representation is a recurring limitation across the literature. Candé and Martinho (2026) identify limited language datasets, inconsistent linguistic standardization, and weak representation in training datasets as major constraints for low resource languages. These factors reduce translation quality and model generalization, particularly in specialized contexts.

Language barriers can also directly affect subject learning. Athukorala and De Silva (2025) demonstrate this through a programming assistance system for Sinhala-speaking Java learners. Allowing learners to express programming problems in Sinhala improved conceptual understanding and reduced difficulties associated with English terminology. Their findings show that language support can facilitate access to technical knowledge rather than merely translate text.

Effective multilingual support must therefore account for both linguistic meaning and subject knowledge. Performance gains achieved for high-resource languages cannot be assumed to transfer uniformly to underrepresented languages because model performance differs according to linguistic representation, resource availability, and domain (Candé & Martinho, 2026; Vatsal et al., 2026). Educational systems for low resource languages should consequently be designed around the characteristics of the target language and learning context.

2.2. LLM-Based Translation and STEM Terminology

Translation is prominent across the reviewed literature, but specialized learning requires more than literal substitution. He et al. (2025) show that translation in cross-cultural online education must preserve instructional meaning and cultural context. Their study also identifies difficulties involving low resource languages, rare vocabulary, and culturally specific expressions.

The challenge is particularly important for technical terminology. Kleidermacher and Zou (2026) demonstrate that LLMs can translate scientific texts competitively while following detailed instructions and in-context examples. Their study also shows that some scientific terms may be better retained in English when direct translation produces unfamiliar or misleading terminology. This suggests that STEM terminology support should allow selective preservation and contextual explanation rather than automatically translating every technical term.

Terminology must also be identified before it can be appropriately translated or explained. Tran et al. (2026) position Automatic Term Extraction as the identification of specialized domain terms rather than ordinary vocabulary. This capability is relevant to an educational artefact because technical concepts need to be distinguished from general language before suitable support can be generated.

Overall, the evidence supports context-aware and terminology-sensitive translation. Because LLM performance remains uneven in specialized contexts, generated outputs also require critical evaluation and appropriate human supervision (Candé & Martinho, 2026).

2.3. Educational Scaffolding and Personalized Learning

The literature demonstrates a shift from using AI primarily for translation towards using it as an educational scaffold. LLMs can support learners through explanations, feedback, multilingual interaction, and adaptive assistance (Candé & Martinho, 2026). Kuzu (2026) demonstrates this in multilingual mathematics activities, where learners formulated prompts, evaluated responses, and refined explanations. This suggests that effective scaffolding requires structured interaction and learner verification rather than unrestricted access to an LLM.

Educational systems provide further evidence of this approach. Rakhimova et al. (2024) combine textual, visual, and audio resources in an educational dictionary, while Zhang and Pang (2025) integrate summaries, multilingual subtitles, and question answering into video learning. Nair et al. (2026) provide bilingual and adaptive support for Telugu learners, whereas Sakunkoo et al. (2025) use code-switched content for contextual vocabulary learning.

Despite their different approaches, these systems move beyond passive translation towards structured and learner-oriented support. However, they provide limited attention to systematically identifying and explaining specialized STEM terminology. This suggests an opportunity to combine multilingual educational scaffolding with terminology-sensitive support for low resource STEM learners.

2.4. System Design and Multilingual Prompt Engineering

Prompt engineering provides a practical method for adapting general-purpose LLMs to multilingual tasks. Vatsal et al. (2026) review multilingual prompting across numerous NLP tasks and demonstrate that effectiveness varies by language, task, and resource availability. Few-shot, bilingual, cross-lingual, and structured prompting can guide model behavior, although more complex prompts do not necessarily produce better results.

Prompting is therefore better treated as a design mechanism that requires empirical evaluation rather than an assumed solution. Bal and Mandal (2026), for example, use hierarchical and adaptive bilingual prompting to improve dialogue processing for Hindi and Bengali without task-specific fine-tuning. Khoboko et al. (2025) similarly demonstrate that carefully designed prompts can improve English-to-Zulu and English-to-Xhosa translation.

Several studies also show that LLMs are most useful when integrated with other system components. Athukorala and De Silva (2025) combine translation, ChatGPT, APIs, and a learner-facing interface in their programming assistant. Zhang and Pang (2025) integrate LLM services into a video-learning platform, while Nair et al. (2026) combine LLM capabilities with speech recognition and bilingual feedback. These studies support treating the LLM as one component within a broader educational information system.

2.5. SLR Theme and Sub-theme Framework

The thematic analysis produced four themes and twelve sub-themes. These capture the principal challenges, educational capabilities, and design mechanisms identified across the reviewed literature.

Table 1
SLR Themes and Sub-themes

Theme	Sub-theme	Description
Low Resource and Multilingual Language Barriers	Low Resource Language Challenges	Limitations arising from scarce datasets, linguistic resources, language technologies, or representation in LLM training.
	Multilingual Learning Barriers	Language-related difficulties that affect learners' access to or understanding of educational content.
	Cultural and Linguistic Context	The need to preserve or account for language-specific and cultural meaning when supporting multilingual learners.
LLM-Based Translation and Terminology Support	Context-Aware Translation	Translation that considers instructional, semantic, or cultural context rather than relying solely on literal substitution.
	Technical Terminology	Translation, preservation, or explanation of specialized scientific and technical terms.
	Terminology Identification	Identification or extraction of domain-specific terms before translation or explanation.
Educational Scaffolding and Personalized Learning	Conceptual Explanation	Explanations that help learners understand concepts rather than only translating words or sentences.
	Personalized/Adaptive Support	Adaptation of explanations, feedback, content, or activities to individual learner needs.
	Multilingual Interaction	Interactive learning in which learners can use or move between multiple languages.
	Structured Learning Support	Guided activities, feedback, question answering, and other mechanisms that structure the learning process.
System Design and Prompt Engineering	Prompt Engineering	Use of structured, bilingual, few-shot, or cross-lingual prompting to guide LLM behavior.

The framework demonstrates that the problem spans linguistic, educational, and technological dimensions. Effective low resource STEM support therefore requires knowledge from all four themes rather than translation capability alone.

2.6. SLR Heatmap

A heatmap was developed to compare the 17 studies across the identified themes and sub-themes. Each study was assessed using a four-level scale: 3 = strong coverage, 2 = moderate coverage, 1 = limited coverage, and 0 = not addressed. Strong coverage indicates that a sub-theme represents a major focus supported through substantial analysis, implementation, evaluation, or synthesis. Moderate coverage indicates meaningful treatment, while limited coverage represents secondary consideration. Scores were assigned based on the extent to which each sub-theme was substantively analyzed, implemented, evaluated, or synthesized in the full text, rather than on the frequency of keyword occurrence. A red color scale was used to visualize coverage, with darker red indicating stronger coverage of a sub-theme.

Table 2

Coverage of SLR Themes and Sub-themes Across Included Studies

Authors	Low Resource and Multilingual Language Barriers			LLM-Based Translation and Terminology Support			Educational Scaffolding and Personalized Learning				System Design and Prompt Engineering	
	Low Resource Language Challenges	Multilingual Learning Barriers	Cultural and Linguistic Context	Context-Aware Translation	Technical Terminology	Terminology Identification	Conceptual Explanation	Personalised / Adaptive Support	Multilingual Interaction	Structured Learning Support	Prompt Engineering	Integrated Educational System
He et al. (2025)	2	2	3	3	1	0	1	0	1	1	1	1
Bal and Mandal (2026)	3	1	3	2	0	0	0	0	1	0	3	0
Kuzu (2026)	1	3	2	1	0	0	3	1	3	3	3	2
Zhang and Pang (2025)	0	1	0	1	0	0	2	1	1	3	1	3
Candé and Martinho (2026)	3	3	3	3	2	0	2	2	2	2	1	1
Athukorala and De Silva (2025)	3	3	1	3	3	1	3	1	2	3	1	3
Yu et al. (2025)	1	3	2	2	1	0	2	3	3	3	1	1
Rakhimova et al. (2024)	3	2	2	2	1	1	2	1	1	3	0	3
Ataman et al. (2025)	3	1	2	3	2	1	0	0	0	0	2	0
Vatsal et al. (2026)	2	1	2	2	1	0	0	0	0	0	3	0
Khoboko et al. (2025)	3	0	1	3	2	1	0	0	0	0	3	0
Tran et al. (2026)	1	0	0	0	3	3	0	0	0	0	1	0
Kleidermacher and Zou (2026)	2	1	1	3	3	1	1	0	1	0	3	0
Sakunkoo et al. (2025)	0	2	2	3	1	0	1	2	3	2	2	3
Bo et al. (2025)	1	2	2	2	1	0	2	3	2	2	1	0
Yüksel et al. (2024)	2	1	1	0	1	0	0	0	0	0	1	0
Nair et al. (2026)	3	3	3	2	0	0	2	3	3	3	3	3

0 = No coverage, 1 = Limited, 2 = Moderate, 3 = Strong; darker red indicates stronger coverage.

The heatmap reveals fragmentation across the literature. Translation-oriented studies provide comparatively strong coverage of low resource language challenges, context-aware translation, and prompt engineering. Scientific translation and term-extraction research also provide knowledge concerning specialized terminology. However, these studies generally provide less coverage of conceptual explanation, personalization, and structured educational scaffolding.

Conversely, educational applications provide stronger evidence for multilingual interaction, conceptual support, personalized assistance, and structured learning activities but give comparatively less attention to identifying and managing specialized STEM terminology. No reviewed study strongly covers all twelve sub-themes.

The evidence therefore suggests that relevant capabilities are distributed across separate research streams. Existing studies provide valuable knowledge about multilingual translation, terminology, educational scaffolding, personalization, and LLM-based system design, but few integrate these capabilities. In particular, none of the reviewed studies demonstrates an integrated approach that combines low resource language support, STEM terminology handling, conceptual explanation, personalized scaffolding, and prompt-engineered interaction. This absence is particularly relevant to the proposed context of Burmese-speaking STEM learners and establishes the basis for the research gap developed in the following SLR synthesis.

3. SLR Synthesis and Research Gap

The SLR reveals that the challenge of supporting STEM learning in a low resource language is not addressed adequately by any single research stream. Instead, relevant knowledge is distributed across four areas: low resource and multilingual language support, LLM-based translation and terminology, educational scaffolding and personalization, and system design and prompt engineering. The heatmap demonstrates that studies tend to provide strong coverage within one or two of these areas but substantially weaker coverage across the others. This fragmentation reveals several interconnected research problems.

First, general multilingual capability does not guarantee effective support for low resource languages. Low resource languages remain constrained by limited linguistic resources, weak representation in training data, and uneven model performance (Candé & Martinho, 2026; Vatsal et al., 2026). Consequently, improvements demonstrated for high-resource languages cannot automatically be assumed to apply to Burmese or other underrepresented languages.

Second, translation alone does not address the conceptual demands of STEM learning. Scientific and educational translation studies demonstrate the importance of context, terminology, and domain knowledge.

Kleidermacher and Zou (2026), for example, show that scientific translation may require specialized prompting and selective preservation of technical terms rather than direct translation. At the same time, Automatic Term Extraction research demonstrates the importance of distinguishing domain-specific terminology from ordinary vocabulary (Tran et al., 2026). The heatmap reinforces this problem: terminology identification is among the least consistently addressed sub-themes across the reviewed studies.

Third, language support and educational scaffolding are insufficiently integrated. Educational studies demonstrate that LLMs can support conceptual explanation, multilingual interaction, feedback, and personalized learning. Kuzu (2026) shows that multilingual ChatGPT interaction can support mathematical problem solving when incorporated into structured learning activities, while Athukorala and De Silva (2025) demonstrate the value of native-language support for programming concepts. However, educational systems generally provide less attention to systematic identification and treatment of specialized STEM terminology.

Finally, technical and pedagogical capabilities remain fragmented across existing system designs. Prompt-engineering research demonstrates methods for controlling multilingual LLM behavior, while educational systems demonstrate the value of integrating LLMs with interfaces and other technologies. Yet the heatmap shows that strong coverage of prompt engineering, terminology handling, personalization, conceptual explanation, and integrated educational system design rarely occurs within the same study.

3.1. Research Gap

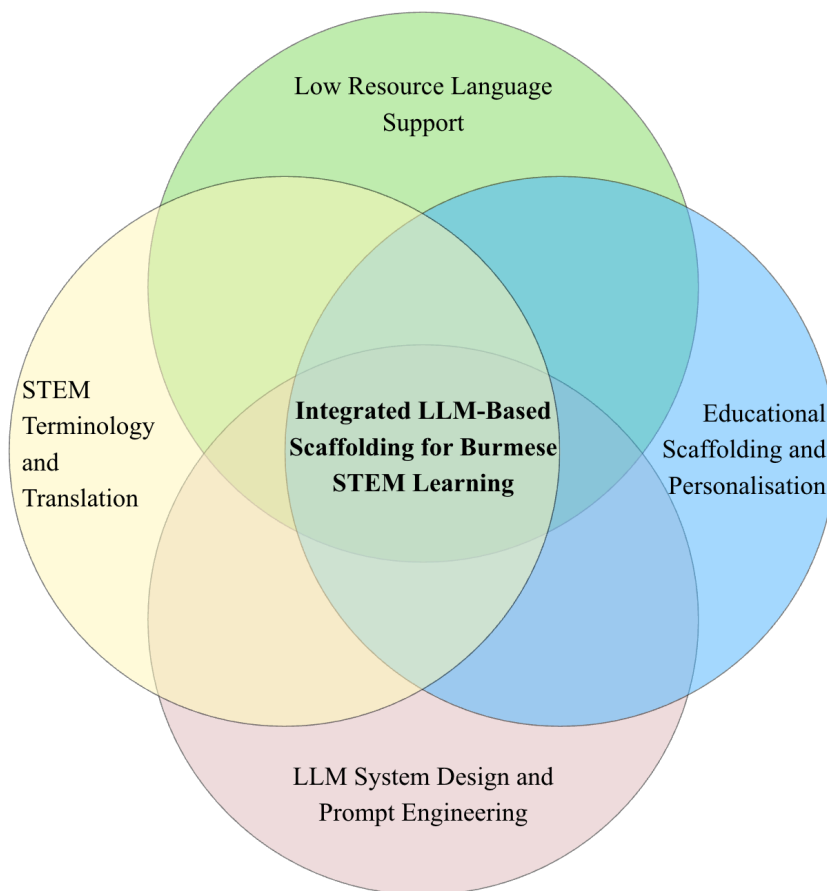
The synthesis identifies a gap at the intersection of low resource language support, STEM terminology support, educational scaffolding, and LLM-based system design. Existing studies demonstrate individual capabilities such as multilingual translation, terminology extraction, conceptual explanation, personalized learning, and prompt engineering, but few combine them into a coherent educational approach. In particular, the reviewed literature does not demonstrate an integrated approach that combines specialized STEM terminology identification, context-aware low resource language support, conceptual explanation, and structured LLM interaction for STEM learning. This absence is particularly relevant to Burmese-speaking learners, who form the target context of this research.

Figure 2 synthesizes these findings by positioning the four research streams identified through the SLR as overlapping areas. Prior studies provide substantial knowledge within individual areas and some intersections; however, the heatmap shows limited integration across all four. The central intersection therefore represents the research gap: integrated LLM-based scaffolding for Burmese STEM learning.

No study in the reviewed sample examined an integrated LLM-based STEM scaffolding system for Burmese-speaking learners. More specifically, the reviewed sample does not demonstrate an integrated educational approach that combines language support with terminology-sensitive and educationally structured assistance to help learners understand specialized STEM concepts. Addressing this intersection provides an opportunity to derive conceptual design knowledge for integrating low resource language support, STEM terminology handling, educational scaffolding, and controlled LLM interaction. These findings provide the foundation for the SSR and the subsequent derivation of research requirements and artefacts.

Figure 2

Venn Diagram of the Research Gap



4. Systematic Systems Review (SSR)

The Systematic Systems Review (SSR) examined implemented systems identified through the reviewed literature to determine how existing technologies address multilingual learning, low resource language support, educational scaffolding, and LLM-supported interaction. While the SLR established the conceptual research gap, the SSR focuses on existing technological artefacts and the design knowledge they provide for the proposed system. This follows the systems-development perspective that existing technologies and systems can provide knowledge for constructing and evaluating new Information Systems artefacts (Nunamaker et al., 1990).

Five systems were selected because they provide implemented functionality closely related to different dimensions of the research problem: native-language STEM assistance, educational terminology resources, LLM-supported learning, multilingual interaction, and personalized language support. Together, they provide complementary evidence for system design while revealing capabilities that remain insufficiently integrated.

Table 3

Systematic Systems Review of Related Systems

System	Functionality / Features / Affordances	Limitations	Gap Relevant to Proposed System
Java Programming Assistance Tool for Sinhala Native Speakers (Athukorala & De Silva, 2025)	Allows learners to formulate Java programming questions in Sinhala. Integrates language translation, ChatGPT, APIs, and a learner-facing interface to provide programming assistance and explanations. Affords native-language access to technical learning and reduces dependence on English interaction.	Restricted to Sinhala and Java programming. The system does not report a separate automatic term-identification or extraction stage before translation.	Closely demonstrates native-language STEM assistance but lacks a general mechanism for identifying and scaffolding specialized terminology across STEM domains.
Children's Educational Dictionary for Kazakh (Rakhimova et al., 2024)	Organizes AI-generated linguistic resources into an educational dictionary with textual, visual, and audio material. Affords multimodal vocabulary learning and reusable structured language resources.	Designed for children's general vocabulary rather than specialized STEM concepts. The implemented system does not report conversational scaffolding or individualized adaptive study plans.	Demonstrates structured presentation of language resources but lacks dynamic, context-aware STEM terminology explanation and learner-specific support.
Interactive Video Learning Framework (Zhang & Pang, 2025)	Integrates LLM-generated summaries, multilingual subtitles, mind maps, and real-time question answering into a learning platform. Affords multiple representations of educational content and interactive support.	Centered on video learning rather than low resource language or specialized terminology. Multilingual support does not explicitly identify or manage STEM terms.	Demonstrates how LLM capabilities can be embedded within structured learning but lacks targeted low resource STEM terminology support.
SingLing (Sankunkoo et al., 2025)	Uses code-switched song lyrics, contextual vocabulary support, and interactive learning activities. Affords contextual multilingual learning and adaptation of content to learners.	Primarily supports general language acquisition. Its learning approach is not designed for technical terminology or conceptual STEM knowledge.	Provides design knowledge for contextual multilingual interaction and personalization but does not address specialized STEM learning.
Vidyālaya (Nair et al., 2026)	Provides Telugu language learning through bilingual interaction, LLM capabilities, speech technologies, personalized feedback, and structured learning activities. Affords interactive and adaptive learning in an underrepresented language context.	Focuses on language acquisition rather than subject-specific STEM learning. Identification and treatment of technical terminology are not central functions.	Demonstrates integration of multilingual interaction, personalization, and AI components, but lacks domain-specific terminology scaffolding.

The comparison reveals three important design insights. First, native-language interaction is technically feasible and educationally useful. The Sinhala programming assistant provides the strongest example because it connects native-language input directly with technical learning. However, its restriction to Java illustrates the need for a more general approach capable of supporting terminology across STEM domains.

Second, existing systems show that translation should be embedded within structured educational functionality. The Kazakh dictionary organizes language resources into multimodal learning material, while the interactive video framework combines multilingual access with summaries, question answering, and other learning mechanisms. SingLing and Vidyālaya similarly demonstrate how contextual interaction and personalization can make multilingual support more educationally meaningful. These affordances suggest that a proposed system should not operate as a simple translation interface or unrestricted chatbot.

Third, the systems reveal a persistent integration gap. No reviewed system combines automatic identification of specialized STEM terminology with context-aware low resource language support, conceptual explanation, personalized scaffolding, and controlled LLM interaction. Systems that provide strong educational scaffolding generally lack specialized terminology processing, while systems addressing technical or linguistic problems provide fewer educational mechanisms.

The SSR therefore provides concrete grounding for the proposed system design. The future artefact should integrate STEM terminology identification, context-sensitive Burmese support, conceptual explanation, appropriate preservation of technical terms, personalized scaffolding, and structured LLM interaction within a learner-facing educational system. These system-level findings complement the SLR research gap and inform the requirements and artefacts articulated through PIRQOA in Section 5.

5. Gap, Kernel Theories and PIRQOA

The combined SLR and SSR identify a gap at the intersection of low resource language support, STEM terminology, educational scaffolding, and LLM-based system design. The SLR showed that translation-oriented research provides knowledge about multilingual translation, terminology, and prompting, while educational research provides knowledge about conceptual explanation, personalization, and structured learning. The SSR further showed that existing systems implement some of these capabilities, but none integrates them comprehensively for low resource STEM learning.

The resulting research problem is therefore not simply the provision of Burmese translation. General translation may provide a target-language equivalent without helping a learner understand the specialized concept. The proposed research addresses the need for an educational approach that can identify STEM terminology, provide context-sensitive Burmese support, explain concepts at an appropriate level, and structure LLM interaction around learning objectives.

5.1. Kernel Theories

Two kernel theories provide justificatory knowledge for the conceptual and system artefacts.

Scaffolding Theory provides the primary educational foundation. Wood et al. (1976) conceptualize tutoring as support that enables a learner to accomplish tasks that would otherwise exceed their current capability. Applied to this research, an LLM should therefore do more than translate terminology: it should provide appropriate explanations, examples, prompts, and progressive assistance that help learners understand STEM concepts.

Cognitive Theory of Multimedia Learning (CTML) provides complementary grounding for presenting explanations. Mayer (2024) explains that meaningful learning from words and visual representations depends on active cognitive processing and is constrained by limited processing capacity. This theory informs the organization and presentation of translated terminology, explanations, examples, and supporting representations so that additional language support does not unnecessarily increase cognitive demands.

Together, these theories provide justificatory knowledge for what educational support the artefact should provide and how that support should be presented. The system artefact itself is situated within Design Science Research, where knowledge about a problem and its solution can be developed through the construction and application of an artefact (Hevner et al., 2004).

5.2. PIRQOA

The SLR, SSR, and kernel theories lead to the following PIRQOA.

Table 4
PIRQOA Derived from the SLR and SSR Findings

Problems	Issues	Requirements	Research Questions	Objectives	Artefacts
Language barriers can make specialized English STEM terminology difficult to understand for low resource language learners.	Insufficient terminology and language support	Identify specialized STEM terminology and provide context-sensitive Burmese support, preserving English technical terms when appropriate.	RQ1: How can specialized English STEM terminology be supported for Burmese-speaking learners?	Develop an approach for multilingual STEM terminology support.	Conceptual Artefact: STEM Terminology Support Framework System Artefact: STEM Terminology Support Prototype
Low resource languages have limited linguistic resources and weaker representation in language technologies.					
Direct translation may produce unfamiliar or misleading technical terms.					
Specialized terminology is not consistently identified					

before translation or explanation.					
Translation alone may not provide enough conceptual support for STEM learning.	Insufficient conceptual learning support	Provide clear explanations and examples beyond translation.	RQ2: How can LLM-based support help learners understand STEM concepts beyond translation?	Develop an approach for conceptual STEM explanation.	Conceptual Artefact: STEM Concept Explanation Framework System Artefact: STEM Concept Explanation Prototype
Translation alone may not provide enough conceptual support for STEM learning.					
Existing educational systems provide limited personalized support for specialized STEM terminology.	Insufficient structured and adaptive scaffolding	Provide structured and adaptive learner support.	RQ3: How can LLM-based scaffolding provide structured and adaptive support for Burmese-speaking STEM learners?	Develop an adaptive scaffolding approach.	Conceptual Artefact: Adaptive STEM Scaffolding Model System Artefact: Adaptive Scaffolding Prototype
Unstructured LLM interaction may provide insufficient educational scaffolding.					

The PIRQOA forms a traceable progression from the identified research problem to the intended artefacts. The SLR establishes the fragmented knowledge base, the SSR identifies missing system capabilities, and the kernel theories justify the proposed educational mechanisms. The conceptual artefacts are positioned before the corresponding system artefacts so that each prototype implements theoretically and empirically grounded design knowledge rather than functioning as an isolated application.

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